

Signal Integrity and Compliance Test of DSI and CSI2 Serial Interface over MIPI D-PHY

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Abstract— In this paper, signal integrity (SI) analysis and compliance test procedure of MIPI D-PHY layer are discussed using a time-domain based simulation technique. Compliance testing is a performance measure for D-PHY to ensure channel parameters are as per MIPI specifications. DSI and CSI2 serial interfaces are analyzed as per design specifications developed by the MIPI PHY working group. Simulated results are validated as per MIPI compliance specifications for various test setup. A frequency-domain full-wave electromagnetic simulator is used to evaluate channel loss. For transient circuit simulation of the channel, IBIS-AMI model of D-PHY transmitter and receiver are incorporated. Channel simulation of MIPI interconnects, including an equalizer and channel components, is carried out to validate the eye characteristics.

Keywords— MIPI, SerDes, SI/PI, CIS2, DSI, IBIS-AMI model, transient simulation, signal integrity analysis, compliance test

I. INTRODUCTION

High-Speed Digital (HSD) designs have several interfaces between the host processor and other connected digital devices [1]. Any smart device in today's era consists of various high-speed interfaces, namely the display, camera, memory, Ethernet, etc., operating at varying data speed. Mobile Industry Processor Interface (MIPI) alliance provides a complete list of specifications to connect chipsets and peripherals in mobile-influenced devices [2]. MIPI is an organization helping industries that develop serial interface connects for mobile and mobile-influenced device. The MIPI D-PHY (physical layer) is used for display and camera interfaces in mobile applications with a short channel, and it follows Serializer/Deserializer (SerDes) architecture. SerDes is a serial data transceiver that first serializes the parallel data into serial data at the transmitter and then deserializes the serial data again to parallel data at the receiver.

SI analysis is performed to test the performance and functionality of the MIPI transmitter and receiver to ensure it is not causing signal impurities. The goal of signal integrity analysis is to optimize channel performance as per MIPI D-PHY specifications. In the rest of the paper, compliance test and signal integrity solution of MIPI D-PHY using simulation technologies are discussed.

II. MIPI PHYSICAL LAYERS

MIPI alliance groups high-speed physical layers in three standards: MIPI D-PHY, M-PHY, and C-PHY. These three different layers serve important interconnection needs in a board or a device. D-PHY and C-PHY standards support

display and camera applications, while the M-PHY layer supports chip-to-chip interfaces, the high-performance memory, and camera. There is one more layer, A-PHY, that is in the progress of development. D-PHY physically connects the camera module to the processor and processor to the display module. It is used in smartphones for displays and cameras because it provides high speed, low power, flexible, and low-cost solution. Display Serial Interface (DSI) and Camera Serial Interface (CSI-2) are the two high-level packet-based protocols that transmit image data between the processor chip and peripheral [3]. As shown in Figure 1, both protocols use MIPI D-PHY at the physical layer.

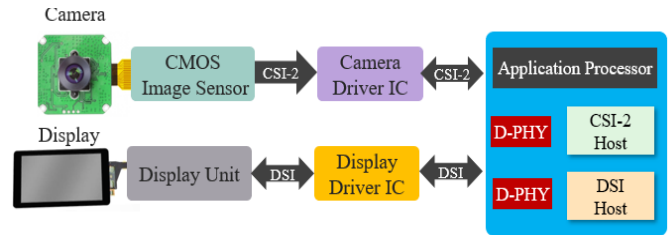


Fig. 1. CSI-2 and DSI interface connections through MIPI D-PHY.

III. MIPI D-PHY ARCHITECTURE

D-PHY uses differential signaling with a source synchronous clock and scalable data lanes. This interface can operate in a duplex or simplex arrangements with multiple data lanes or a single data lane. It uses two interconnect wires per data lane and two wires for the clock signal; therefore, at least four wires are needed for primary minimum connection. Every data lane of the transmitter is connected to the receiver through two wires, Dp and Dn, and clock through two wires Clkp and Clkn. The data signal can be either bi-directional or unidirectional. The clock signal is unidirectional that starts from the transmitter and terminates at the receiver. D-PHY can switch between two operating modes: the single-ended low-power (LP) and differential high-speed (HS). Switching happens in real-time, depending on the requirement to transfer vast amounts of data or to preserve the battery power. LP mode saves energy by raising signal voltage levels to approximately 1.2V and reducing the current and it drives each line of a lane independently. A D-PHY device in HS mode can have any data rate from 80 Mbps to 4.5 Gbps with a low-swing differential signal. LP mode supports 10 Mbps asynchronous data transfer [4]. The architecture of a D-PHY standard is shown in Figure 2.

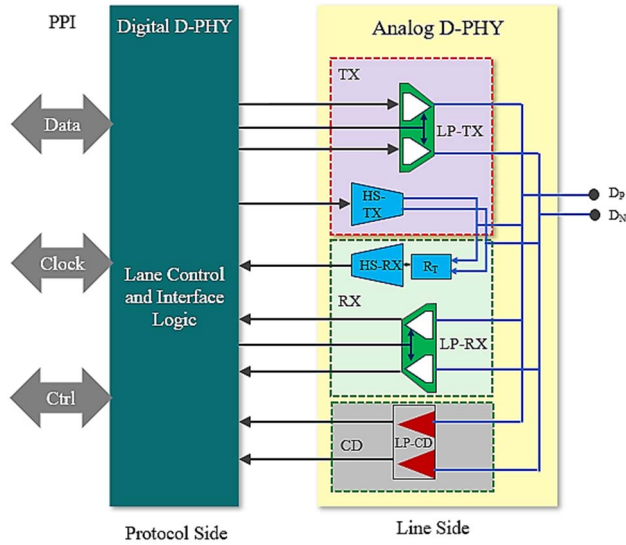


Fig. 2. MIPI D-PHY DSI TX/RX block diagram.

III. ELECTRICAL AND ELECTROMAGNETIC SIMULATION

Circuit simulation on a MIPI serial interface needs circuit models of the transmission line, die, package, vias, transmitter (TX), receiver (RX), and interface connectors. These passive components from the transmitter to the receiver formed the channel.

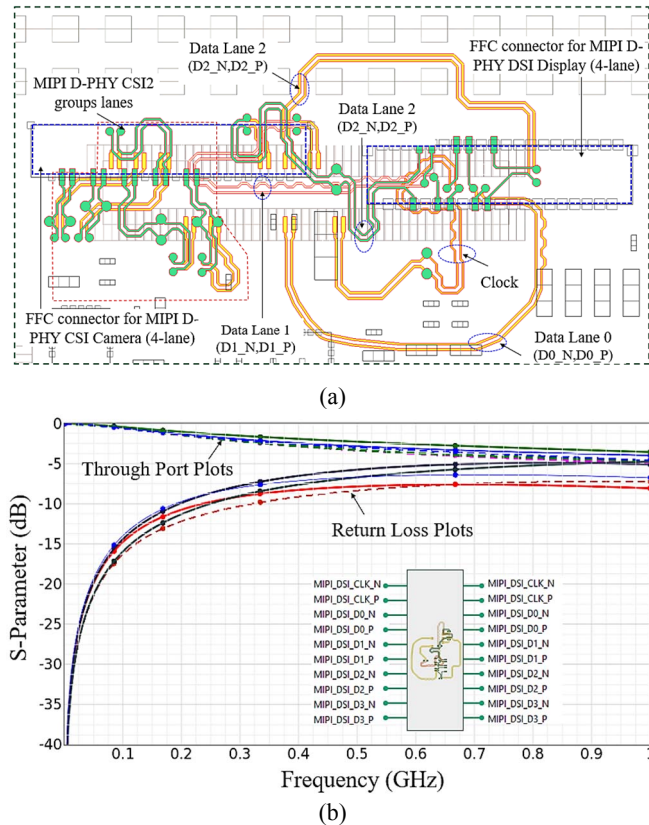


Fig. 3. EM data extraction (a) layout of MIPI D-PHY channel (b) S-Parameter plots of the clock and 4 Lane data channels.

As shown in Figure 3(a), the HSD board with the D-PHY interface, discussed in this paper, is an 8-layer board with 1.2–2.5V power distribution networks (PDN) supplying power to MIPI controller. In the first phase of the analysis process, CSI2 and DSI signal groups are simulated using full-wave Finite Element Method (FEM) based electromagnetic solver. The extracted electromagnetic data, as shown in Figure 3(b), of DSI and CSI2 lane groups are used in the transient simulation for compliance tests and signal integrity. Signal Integrity and Compliance Test OF D-PHY The IBIS-AMI 5.0 transmitter and receiver models are used to analyze different channel parameters. Eye diagrams at the receiver end are compared with eye mask as per the MIPI D-PHY requirement. For good BER (typically 12-17), the simulated eye should be within the eye-opening specification for amplitude and jitter. The equalization for transmitters and receivers is used to reduce inter-symbol interference noise due to channel insertion loss in the high-speed chip-to-chip interface. Compliance testing is performed for both HS and LP mode transmitter and receiver.

(a) HS Mode Transmitter Test

In HS-mode, each lane is driven by a low-swing differential signal and terminated at Rx, as shown in Figure 4. In LP mode, all wires operate as non-terminated and single-ended. The performance of the transmitter, receiver, and interconnect determine the maximum achievable bit-rate. In this mode, the data rate extends up to 2.5 Gbps. The differential signal voltage is 140 mV minimum, 200 mV nominal, and 270 mV maximum. Differential-0 (HS-0) and Differential-1 (HS-1) are two possible states in HS mode.

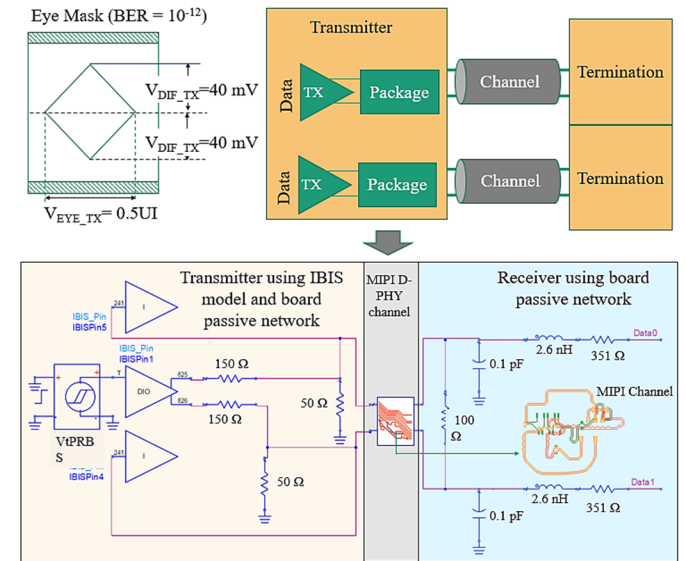


Fig. 4. High-Speed transmitter using the IBIS model.

In HS mode, D-PHY technology utilizes a 100Ω dynamic, resistive termination at the receiver. This resistive termination is enabled in HS mode and disabled (open termination environment) during the LP mode. HS-TX test measures the eye height parameter of the test data signal by generating an

eye diagram based on the data and clock signal, as shown in Figure 5.

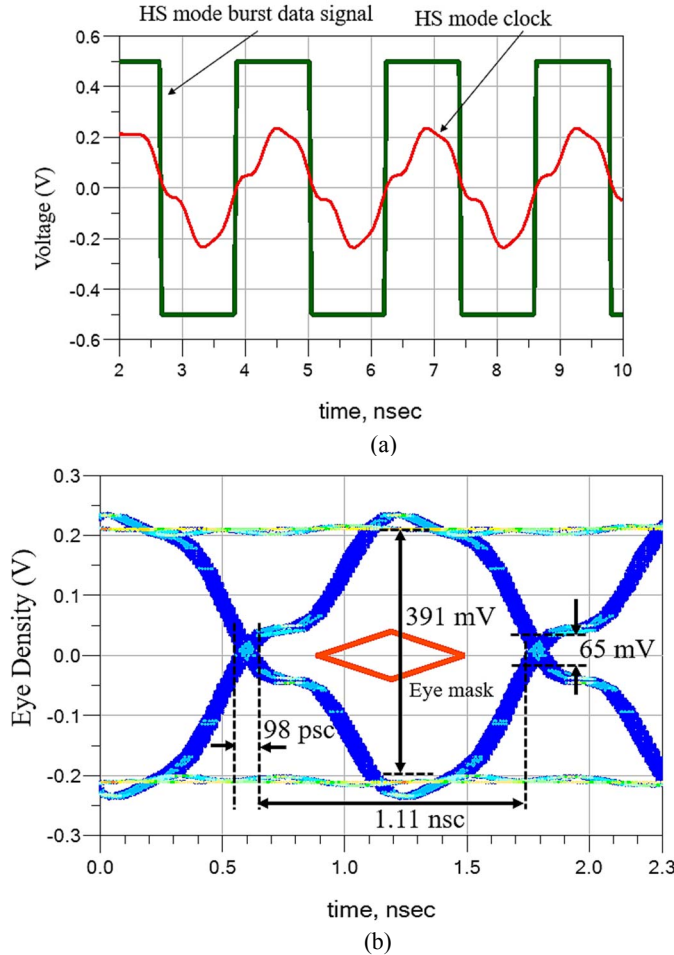


Fig. 5. High-speed transmitter analysis results at @840 Mbps (a) clock signal and burst data (b) eye diagram of the data lane.

The measured eye height must be within the limit as per MIPI D-PHY eye mask. The eye diagram test is performed after applying the TX EQ (de-emphasis). This specification is valid for a data rate from 80 to 1500 Mbps (per lane) without deskew calibration, up to 2500 Mbps with deskew, and up to 4.5 Gbps with equalization. For data rate above 2.5 Gbps, PHY supports Spread Spectrum Clocking (SSC). SSC support on the clock to mitigate emissions concerning electromagnetic interference (EMI) compliance. It is a common technique where a low-frequency modulation is added to the Transmitter's clock to reduce peak emissions. HS data transmission occurs in bursts.

(b) LP Mode Transmitter Test

LP mode transmitter simulation setup and analysis results are shown in Figure 6. LP-TX SI analysis and testing are completed as per the LP mode compliance test specification. The LP transmitter is a slew-rate controlled push-pull driver. LP mode has a 1.2V single-ended signal with a maximum data rate of 10 Mbps. In the D-PHY bus, rise times are within 150ps and 0.3-unit intervals (UI).

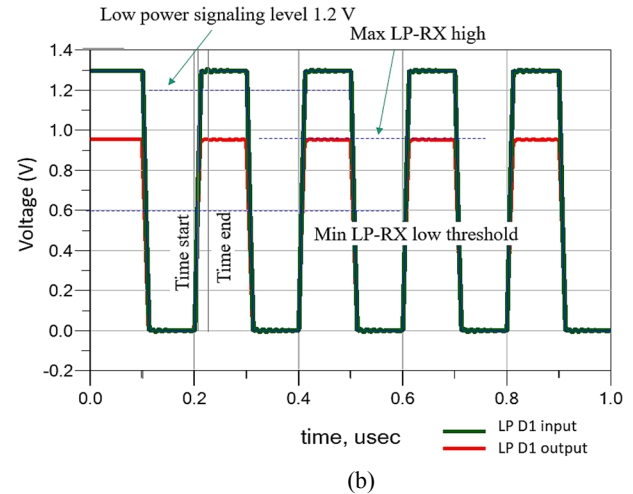
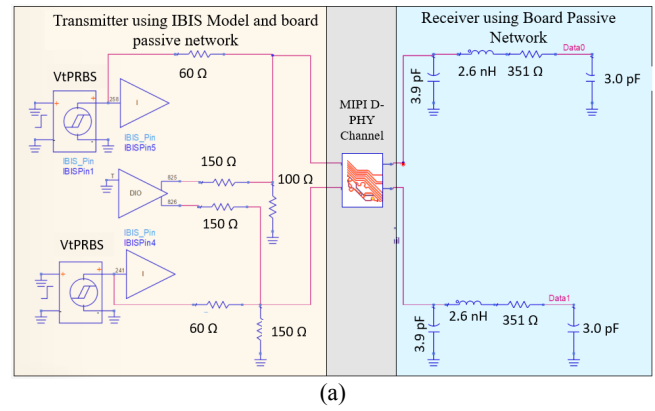
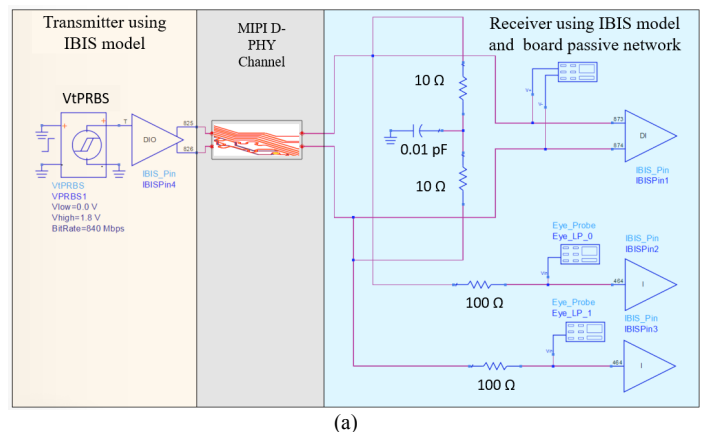


Fig. 6. LP transmitter test (a) test setup (b) Dp and Dn driven in-phase for LP-00, and LP-11

(c) HS Mode Receiver Test

The receiver test is carried out to determine the receiver's functionality and performance to detect the digital signal with worst-case impaired input signals accurately. At the receiver, the signal is measured to determine performance as per target BER. Simulation setup for HS mode receiver testing using IBIS-AMI model is shown in Figure 7(a). The quality of the signal, especially the rise and fall time, amount of ISI, noise, and jitter, are estimated from the appearance of the eye obtained, as shown in Figure 7(b).



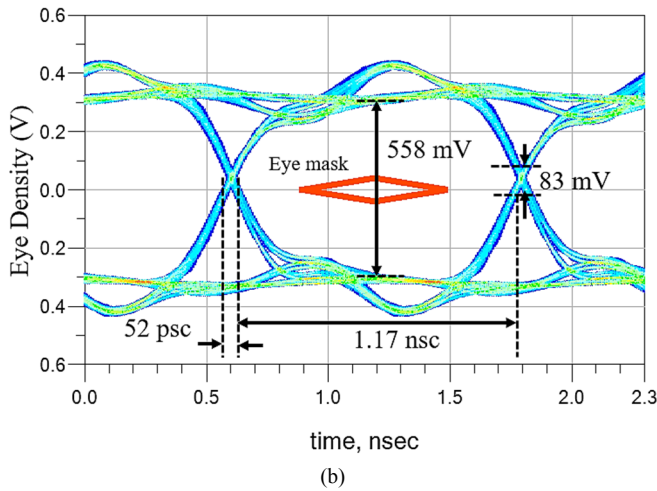


Fig. 7 HS mode receiver testing (a) simulation setup using IBIS-AMI model (b) simulated eye diagram of HS receiver at 2.5 Gbps/lane.

(d) LP Mode Receiver Test

In HS mode, the receiver device presents a 100Ω differential termination while this 100Ω termination is switched to a high impedance (normally 300Ω) when the interface is in LP mode. A high impedance termination avoids the difficulty of switching in and out of the HS mode termination. Signal waveform when Dp and Dn have driven in phase for LP-00 and LP-11, is shown in Figure 8.

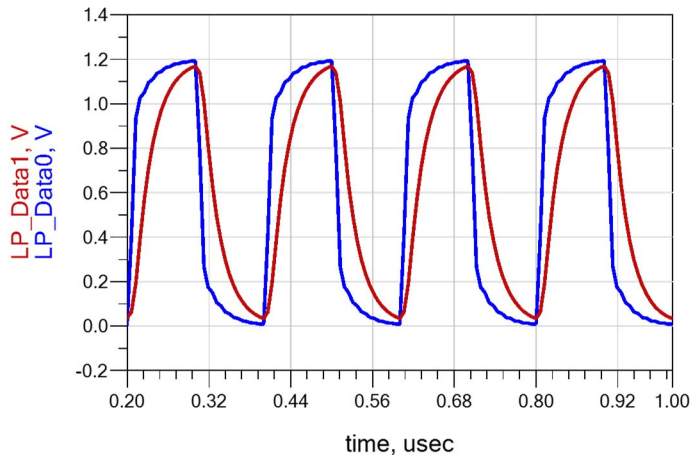


Fig. 8. Dp and Dn have driven in phase for LP-00 and LP-11.

IV. CHANNEL SIMULATION RESULT

For SI analysis on the complete channel, the FFC connector, four-lane data bus, flex cable, and package are first simulated using an electromagnetic solver, as shown in Figure 9. Essential parameters are calculated, including attenuation, reflection, impedance mismatch, crosstalk, propagation delay, flex cable, and the alignment of the connectors are computed by using this electromagnetic S-parameter data. This analysis approach predicts full channel performance. Both flex cable and FFC connectors are included in the analysis.

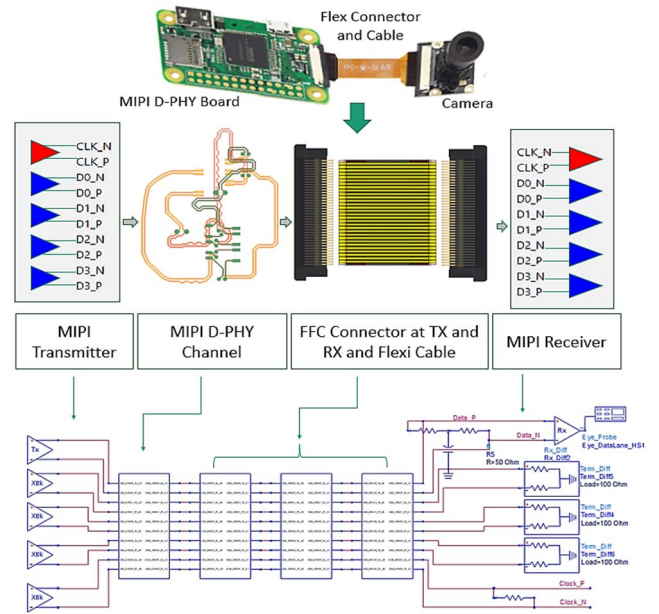


Fig. 9. Simulation setup for the complete channel.

The eye is degraded in a long channel because of the inter-symbol interference. The simulated eye diagram is shown in Figure 10.

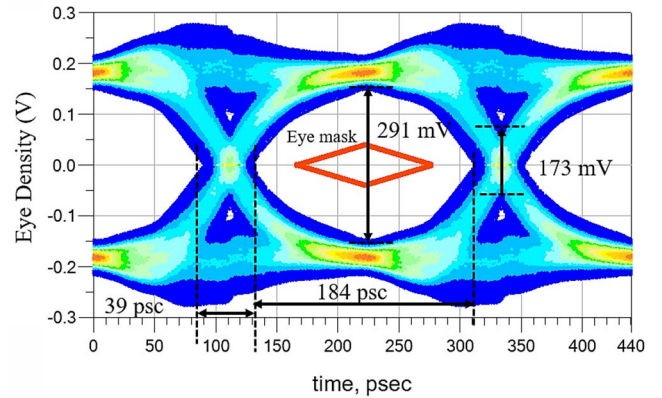


Fig. 10. Simulated eye diagram of the complete channel including the FFC connector, four-lane data bus, flex cable, and package.

V. CONCLUSION

Time-domain signal integrity analysis and compliance test procedure of complete channel of MIPI D-PHY serial interfaces DSI and CSI2 data bus are presented. ADS transient and channel simulators are used to validate the complete D-PHY channel as per MIPI specifications. Compliances testing of TX and RX electrical, as well as complete channel, is done to ensure MIPI channel performance is meeting D-PHY specifications.

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